

- D2850, "Standard Test Method for Unconsolidated-Undrained Triaxial Compression Test on Cohesive Soils"
- D4767, "Standard Test Method for Consolidated Undrained Triaxial Compression Test for Cohesive Soils"

The ASTM standards also described other specialized tests that permit evaluation of the low-strain and dynamic behavior of soils, application of the vane-shear on saturated soils in the laboratory, and evaluation of shear and tensile response of rocks. These ASTM standard tests include the following:

- D5311, "Standard Test Method for Load Controlled Cyclic Triaxial Strength of Soil"
- D3999, "Standard Test Methods for the Determination of the Modulus and Damping Properties of Soils Using the Cyclic Triaxial Apparatus"
- D4015, "Standard Test Methods for Modulus and Damping of Soils by the Resonant-Column Method"
- D4648, "Standard Test Method for Laboratory Miniature Vane Shear Test for Saturated Fine-Grained Clayey Soil"
- D2936, "Standard Test Method for Direct Tensile Strength of Intact Rock Core Specimens"
- D3967, "Standard Test Method for Splitting Tensile Strength of Intact Rock Core Specimens"
- D5607, "Standard Test Method for Performing Laboratory Direct Shear Strength Tests of Rock Specimens under Constant Normal Force"

6.1 Direct shear test

The direct shear test is used to determine the shear strength of soils on a predetermined failure surface. This test is used to measure the friction angle, undrained shear strength, and dilative and contractive tendencies of soils. This test can be conducted in both coarse (sand) soils and fine (clays) soils (Bardet 1997; Lambe 1951). Results of this test are applicable to the stability analysis of foundations, slopes, and retaining walls. The main problems of the test are that (1) the failure plane is forced along an arbitrary plane, (2) the states of the stress in the vertical boundaries are not known (there is a rotation of principal stresses during the test), (3) full saturation cannot be confirmed, and (4) the dissipation of excess pore water pressure is not monitored. In spite of these limitations, the test is quite popular because it is simple to run and the data are easily reduced. This test is described in ASTM standard D3080, "Standard Test Method for Direct Shear Test of Soils Under Consolidated Drained Conditions."

6.1.1 Introduction

Either a square or cylindrical, disturbed or undisturbed soil specimen is confined inside an upper and a lower rigid box and is subjected to the normal load N and the shear force T , while both the horizontal and vertical displacements are measured using dial gauges or linear variable displacement transducers (LVDTs) (Figure 6.1).

Measured parameters include the normal force N , the tangential force T , the vertical displacement δ_v and the horizontal displacement δ_h . Other determined parameters include the vertical stress σ_v (normal force N divided by the specimen cross-sectional area) and the shear stress τ (tangential [shear] force T divided by the specimen cross-sectional area).

The shear strength of the soil specimen is the shear stress τ that causes the soil to slip on the failure surface with normal effective stress σ' , and it can be defined by the Mohr–Coulomb failure criterion:

$$\tau = \sigma' \cdot \tan(\phi) \quad (6.1)$$

where ϕ is the friction angle that represents the effects of interparticle friction and interlocking. Alternatively, Equation 6.1 is presented as

$$\tau = c_a + \sigma' \cdot \tan(\phi) \quad (6.2)$$

where c_a is the shear strength intercept — a model parameter that does not indicate material behavior unless the test is performed in a cemented soil

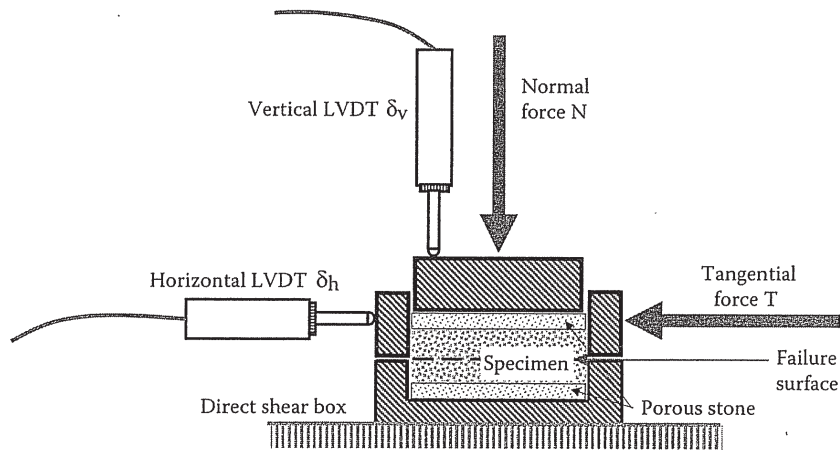


Figure 6.1 Cross-sectional diagram of the direct shear apparatus.

specimen. Performing the direct shear test on soils under different vertical stresses σ_v permits both the c_u and ϕ parameters to be determined.

Equipment

- Direct shear loading device (Figure 6.2): This device must permit (a) the application of constant forces normal to the horizontal faces of the specimen, (b) the application and measurement of the relative horizontal displacement and shear force across arbitrary failure planes parallel to the faces of the specimen, (c) the measurement of the changes in the height of the specimen, (d) submerging the soil specimen during testing, and (e) draining the soil specimen during testing. The horizontal displacement must be applied at a constant rate (0.025 to 1.0 mm/min).
- Direct shear box (Figure 6.3) with either circular to square cross section: The box that is divided in two equal sections along a horizontal plane should be made of bronze, aluminum, or steel. The box

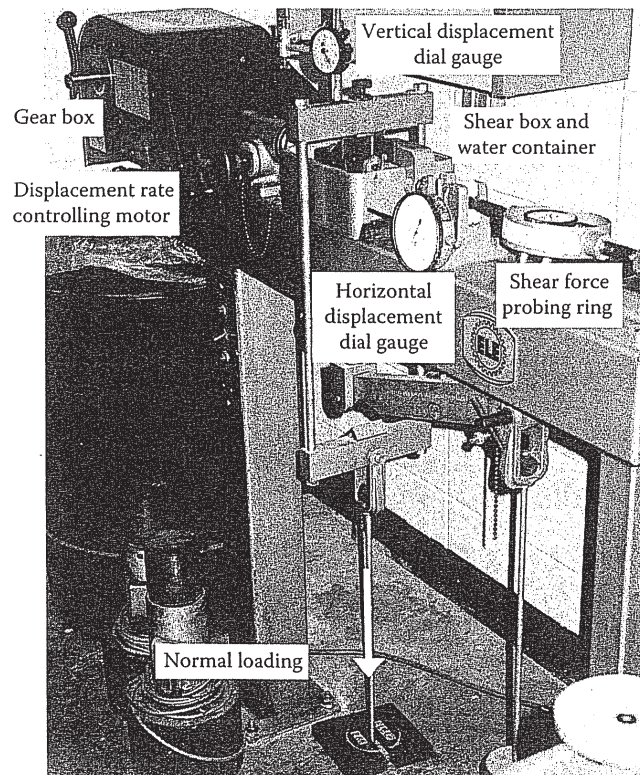


Figure 6.2 Direct shear loading device with analog transducers.

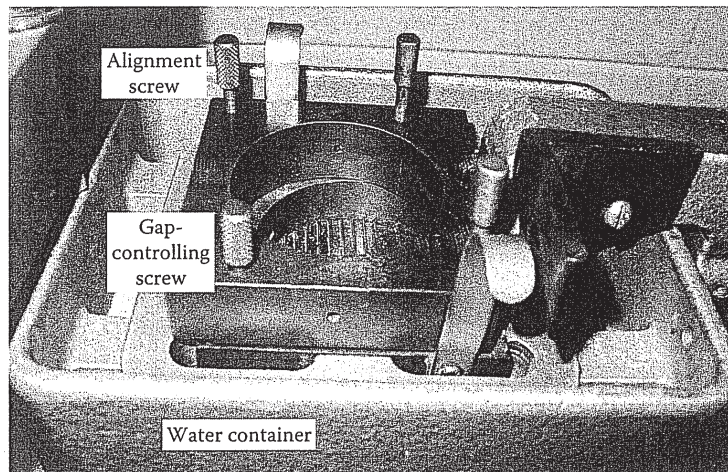


Figure 6.3 Direct shear box. View of the two sections of the circular box and box for submerging the specimen.

should allow the free drainage of water in and out of the soil specimen. The two sections of the box are aligned with screws that need to be removed after specimen preparation and before the beginning of the test. Two other screws are provided to control the gap between the two sections of the box during shear testing.

- Dial indicators: Vertical dial gauge or LVDT (sensitivity 0.0025 mm), horizontal dial gauge or LVDT (sensitivity 0.025 mm), proving ring dial gauge or load cell to measure the applied shear force (sensitivity 2.5 N).
- Balance (0.1 g precision).

Specimen

Undisturbed or disturbed specimen of fine-grained soils (clays) or coarse-grained soils (sand): The maximum particle size of the specimen is controlled by the size of the shear box. (The maximum grain size should be ten times smaller than the cross-sectional dimension of the specimen and six times smaller than the height of the specimen.) The specimen should have a minimum height of 12 mm and a minimum aspect ratio (diameter/length to height) of two.

6.1.2 Procedure

The description of the test will be based on the preparation of remolded sand specimens on a 65 mm diameter shear box. (For details on the preparation

of compacted and undisturbed specimens, refer to ASTM standard D3080.) Three tests will be conducted at three different vertical (normal) loads:

1. Take 130 g of air-dry sand to prepare the specimen.
2. Measure the internal diameter (D) in circular shear boxes or the length (L) in square shear boxes.
3. Measure the mass of the cap, ball bearing, and load hanger ($M_{cap+ball+hanger}$).
4. Assemble the direct shear box and mount it in the direct shear machine (Figure 6.3).
5. Place the sand inside the direct shear box so that the surface of the specimen coincides with the two dot marks inside the box (Figure 6.4). Place the material loosely for the loose specimens. Compact the specimen to obtain dense specimens.

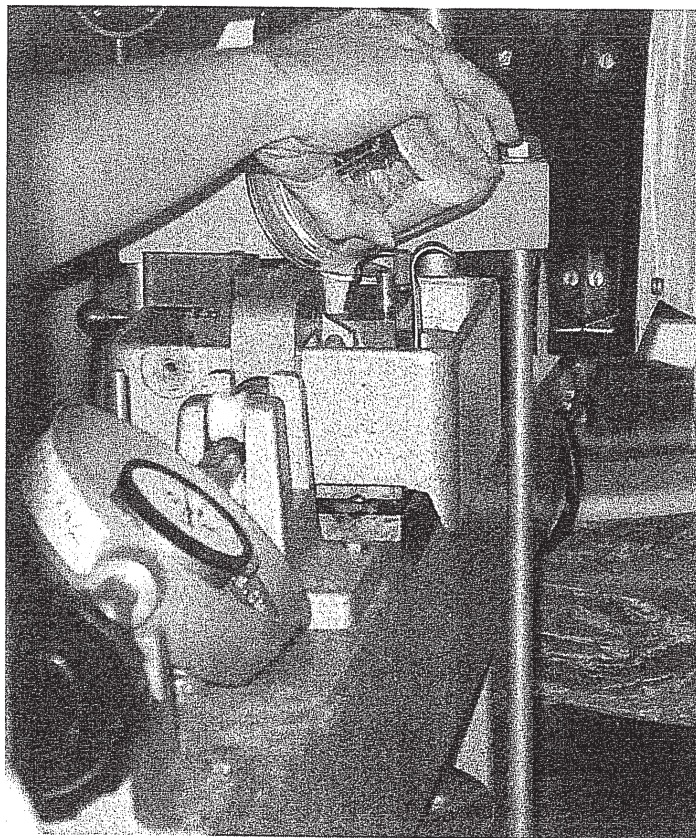


Figure 6.4 Pouring sand into the direct shear box.

6. Determine the mass of the soil specimen (M_s). You can accomplish this by measuring the mass of sand left from the 130 g you used to prepare the specimen.
7. Level the surface inside the shear box.
8. Measure the height of the specimen (H_{specimen}) and calculate the unit weight of the specimen (γ).
9. Assuming the specific gravity is $G_s = 2.65$, calculate the initial void ratio (see Chapter 2).
10. Attach the dial gauges and measure their initial readings.
11. Check that there are no pins left in the shear box.
12. Set the displacement rate at 0.20 mm/min.
13. Apply the normal load (N).
14. Start the loading procedure.
15. The readings of the vertical dial gauge and proving ring are to be recorded for every 0.02 mm of horizontal displacement. (This reading can be performed automatically if an LVDT and a computer-controlled data acquisition system are available.)
16. Stop taking readings when the proving ring shows almost constant readings or after reaching a horizontal strain of 20%.
17. Repeat the test for another soil specimen at a higher normal load.

6.1.3 Data reduction

The following parameters are calculated to interpret the results of the direct shear test (see also Bardet 1997):

Shear stress:

$$\tau = \frac{T}{A_c} \quad (6.3)$$

Vertical stress:

$$\sigma_v = \frac{N}{A_c} \quad (6.4)$$

Corrected area:

$$A_c = \frac{D^2}{2} \cdot [\theta - \cos(\theta) \cdot \sin(\theta)] \quad (\text{Cylindrical cells}) \quad (6.5)$$

where

$$\theta = \cos^{-1} \left(\frac{\delta_h}{D} \right) \quad (\text{in radians}) \quad (6.6)$$

$$A_c = L \cdot (L - \delta_h) \quad (\text{Square cells}) \quad (6.7)$$

Specimen diameter: D

Lateral displacement: δ_h

Vertical displacement: δ_v

Mobilized friction angle:

$$\phi_{mobilized} = \tan^{-1} \left(\frac{\tau}{\sigma_v} \right) \quad (6.8)$$

Peak friction angle:

$$\phi_{peak} = \tan^{-1} \left(\frac{\tau_{max}}{\sigma_v} \right) \quad (6.9)$$

Residual friction angle:

$$\phi_{residual} = \tan^{-1} \left(\frac{\tau_{residual}}{\sigma_v} \right) \quad (6.10)$$

6.1.4 Presentation of results

- Plot $\arctan(\tau/\sigma_v)$ versus horizontal displacement δ_h or horizontal strain $\epsilon_h = \delta_h/D$. (Note that the ratio τ/σ_v is identical to the ratio T/N .) The $\arctan(\tau/\sigma_v)$ is also known as the mobilized friction angle ($\phi_{mobilized}$).
- Plot τ versus horizontal strain ϵ_h for each of the vertical stresses σ_v (Figure 6.5a).
- Plot vertical displacement δ_v versus horizontal strain ϵ_h (Figure 6.5b). This plot indicates the contractive and dilative tendencies of soils and helps determine the critical void ratio. (Please note that the failure plane is forced along a predetermined zone, and the critical void ratio is masked in a soil mass as it does not deform uniformly.)

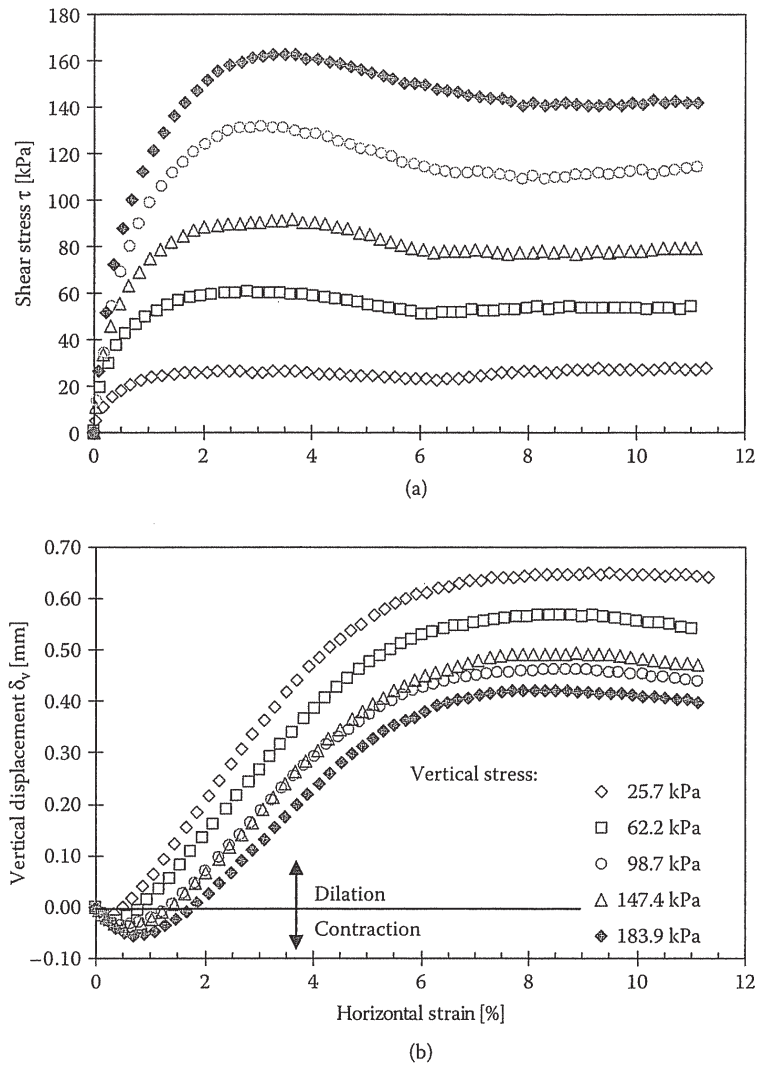


Figure 6.5 Typical test results for a well-graded, dense sand specimen and five different normal effective stresses. (a) Shear stress versus horizontal strain; and (b) vertical displacement versus horizontal strain (data: C. Bareither, University of Wisconsin–Madison). All dense specimens dilate upon shear; however, the specimens have an initial tendency to contract when the vertical stress increases.

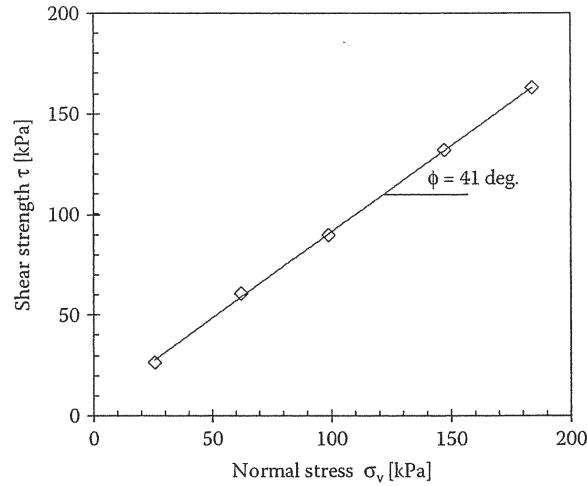


Figure 6.6 Failure envelope for sand specimens at the same initial high relative density.

- Plot τ/σ_v by using T_{max} and $T_{residual}$ values for all vertical stresses σ_v . You will have two best-fit lines: one for maximum stress condition and one for residual stress condition of the sample. The equations for these lines are

$$\tau = c_a + \sigma' \cdot \tan(\phi) \quad (6.11)$$

- The shear intercept (c_a) is 0 kPa for noncemented specimens. The slope of the lines gives the maximum and residual friction angle of the sample. Show these angles on the graph and calculate the ϕ_{peak} and $\phi_{residual}$ values (Figure 6.6).
- Some overconsolidated clay and dense sand specimens may also show a shear intercept (c_a). However, it is important to understand that this shear strength parameter is not a material property but rather the effect of using a linear model (Equation 6.11) in the definition of the failure line that in soils is not a straight line (see also Figure 6.7c).

6.1.5 Sample data and calculations

An example of a direct shear test on a loose sand specimen is presented in Figure 6.7 and Table 6.1.

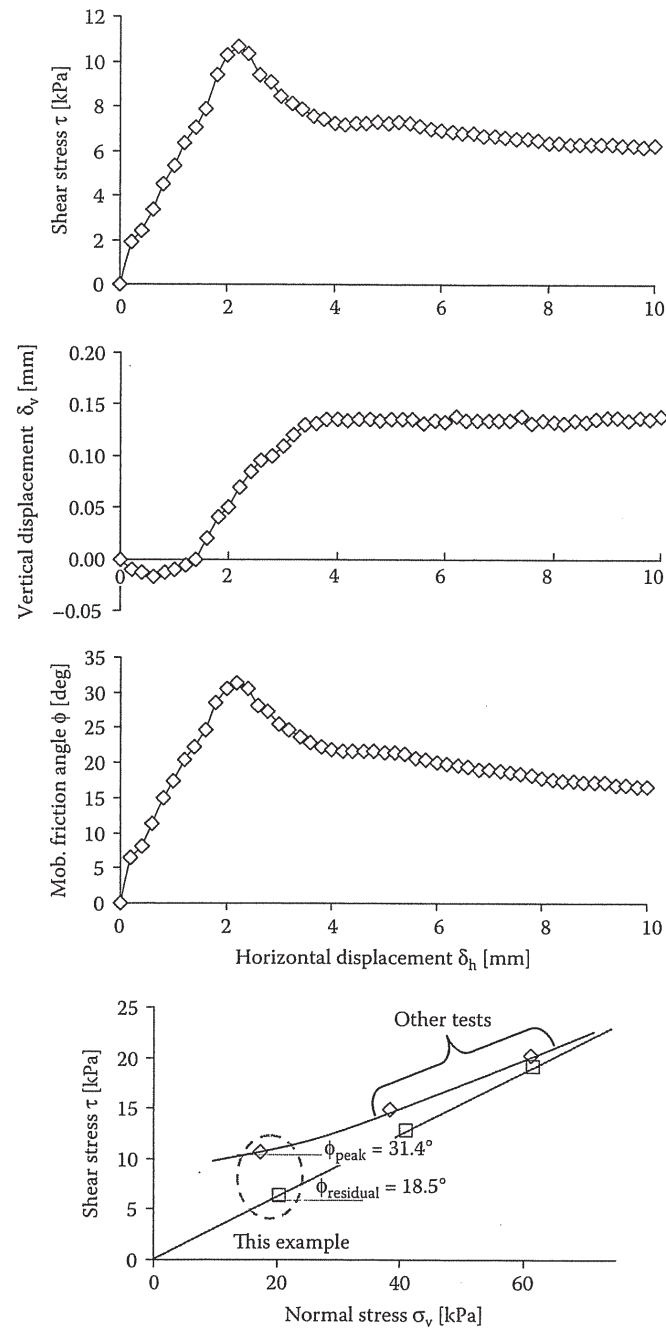


Figure 6.7 Example: Reduction of direct shear data.

Table 6.1 Direct Shear Test: Sample Data

Loose, Clean Sand Specimen

Normal force	$N = 52.42 \text{ N}$ (hanger only)
Specimen mass	$M = 0.15575 \text{ kg}$
Specimen diameter	$D = 0.063 \text{ m}$
Proving ring constant	$K = 50 \text{ N/mm}$
Specimen height	$H = 28.72 \text{ mm}$
Specimen area	$A = 3145.41 \text{ mm}^2$
Initial void ratio	$e_o = 0.535$

Horizontal displace- ment, δ_h (mm)	Vertical displace- ment, δ_v (mm)	Shear force, T (mm)	Shear force T (N)	θ (rad)	A_c (m ²)	σ_v (kPa)	τ (kPa)	T/N
0.000	0.000	52.42	0.00	1.57	0.00315	16.66	0.00	0.00
0.200	-0.010	52.42	6.00	1.57	0.00313	16.72	1.91	0.11
0.400	-0.013	52.42	7.50	1.56	0.00312	16.79	2.40	0.14
0.600	-0.017	52.42	10.50	1.56	0.00311	16.86	3.38	0.20
0.800	-0.013	52.42	14.00	1.56	0.00310	16.93	4.52	0.27
1.000	-0.010	52.42	16.50	1.55	0.00308	17.00	5.35	0.31
1.200	-0.007	52.42	19.50	1.55	0.00307	17.07	6.35	0.37
1.400	0.000	52.42	21.50	1.55	0.00306	17.14	7.03	0.41
1.600	0.020	52.42	24.00	1.55	0.00305	17.21	7.88	0.46
1.800	0.040	52.42	28.50	1.54	0.00303	17.28	9.40	0.54
2.000	0.050	52.42	31.00	1.54	0.00302	17.36	10.26	0.59
2.200	0.070	52.42	32.00	1.54	0.00301	17.43	10.64	0.61
2.400	0.085	52.42	31.00	1.53	0.00300	17.50	10.35	0.59
2.600	0.095	52.42	28.00	1.53	0.00298	17.58	9.39	0.53
2.800	0.100	52.42	27.00	1.53	0.00297	17.65	9.09	0.52
3.000	0.110	52.42	25.00	1.52	0.00296	17.73	8.45	0.48
3.200	0.120	52.42	24.00	1.52	0.00294	17.80	8.15	0.46
3.400	0.130	52.42	23.00	1.52	0.00293	17.88	7.84	0.44
3.600	0.132	52.42	22.00	1.51	0.00292	17.96	7.54	0.42
3.800	0.135	52.42	21.50	1.51	0.00291	18.03	7.40	0.41
4.000	0.135	52.42	21.00	1.51	0.00289	18.11	7.26	0.40
4.200	0.134	52.42	20.75	1.50	0.00288	18.19	7.20	0.40
4.400	0.136	52.42	20.75	1.50	0.00287	18.27	7.23	0.40
4.600	0.135	52.42	20.75	1.50	0.00286	18.35	7.27	0.40
4.800	0.134	52.42	20.75	1.49	0.00284	18.44	7.30	0.40
5.000	0.136	52.42	20.50	1.49	0.00283	18.52	7.24	0.39
5.200	0.136	52.42	20.50	1.49	0.00282	18.60	7.27	0.39
5.400	0.135	52.42	20.30	1.49	0.00281	18.68	7.24	0.39
5.600	0.137	52.42	19.85	1.48	0.00279	18.77	7.11	0.38
5.800	0.132	52.42	19.49	1.48	0.00278	18.85	7.01	0.37
6.000	0.138	52.42	19.27	1.48	0.00277	18.94	6.96	0.37

(Continued)

Table 6.1 Direct Shear Test: Sample Data (Continued)

Horizontal displace- ment, δ_h (mm)	Vertical displace- ment, δ_v (mm)	Shear force, T (mm)	Shear force T (N)	θ (rad)	A_c (m ²)	σ_v (kPa)	τ (kPa)	T/N
6.200	0.138	52.42	19.21	1.47	0.00276	19.03	6.97	0.37
6.400	0.132	52.42	18.84	1.47	0.00274	19.11	6.87	0.36
6.600	0.138	52.42	18.84	1.47	0.00273	19.20	6.90	0.36
6.800	0.136	52.42	18.42	1.46	0.00272	19.29	6.78	0.35
7.000	0.132	52.42	18.09	1.46	0.00270	19.38	6.69	0.35
7.200	0.137	52.42	17.70	1.46	0.00269	19.47	6.57	0.34
7.400	0.134	52.42	17.49	1.45	0.00268	19.56	6.53	0.33
7.600	0.136	52.42	17.45	1.45	0.00267	19.65	6.54	0.33
7.800	0.137	52.42	17.35	1.45	0.00265	19.75	6.54	0.33
8.000	0.137	52.42	17.24	1.44	0.00264	19.84	6.52	0.33
8.200	0.137	52.42	17.02	1.44	0.00263	19.94	6.47	0.32
8.400	0.135	52.42	16.95	1.44	0.00262	20.03	6.48	0.32
8.600	0.133	52.42	16.88	1.43	0.00260	20.13	6.48	0.32
8.800	0.133	52.42	16.82	1.43	0.00259	20.23	6.49	0.32
9.000	0.135	52.42	16.65	1.43	0.00258	20.32	6.46	0.32
9.200	0.133	52.42	16.45	1.42	0.00257	20.42	6.41	0.31
9.400	0.133	52.42	16.35	1.42	0.00255	20.52	6.40	0.31
9.600	0.133	52.42	16.30	1.42	0.00254	20.62	6.41	0.31
9.800	0.135	52.42	16.15	1.42	0.00253	20.73	6.38	0.31
10.000	0.133	52.42	16.07	1.41	0.00252	20.83	6.39	0.31

Questions

- 6.1 Please note that as the soil becomes more contractive, it shows a higher peak in the shear stress horizontal strain response (see Figure 6.5a and Figure 6.5b). Why do you think that is?
- 6.2 Discuss limitations of the direct shear test.
- 6.3 Review Mohr's circle. Why do the directions of principal stress rotate during the direct shear test?

6.2 Unconfined compression test

6.2.1 Introduction

The unconfined compression test is used to measure the unconfined compressive strength of fine-grained soils. This test is applicable only to fine soils such as saturated and unsaturated clays or cemented soils that have shear strength without confining pressure. Clean sands cannot be tested using the unconfined compression test as they cannot keep their form without confining pressure. This test finds its application in the rapid determination of the undrained shear strength of fine-grained soils. The test is described in ASTM standard D2166, "Standard Test Method for Unconfined Compressive Strength of Cohesive Soil."